

GUBBALA ROOP KUMAR

AI/ML Engineer | Agentic AI, RAG & Production LLM Systems

+91 9390737034 | kumargubbala94@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI/ML Engineer specializing in architecting and shipping production-grade Agentic AI, RAG, and LLM applications with autonomous decision-making, combining traditional ML with Generative AI across GCP and AWS (Bedrock, SageMaker). Python-first, with hands-on experience building and integrating LLMs (Gemini, GPT) into production RAG pipelines backed by vector-database retrieval, multi-agent orchestration, and cloud AI services, while developing the guardrails, evaluation loops, API/tool integrations and CI/CD discipline production GenAI systems demand. Proven track record of end-to-end ownership, from data pipeline design through secure, monitored deployment of enterprise AI Systems.

EXPERIENCE

Associate Engineer

Virtusa Consulting Services Private Limited

Jan 2024 – Present

Hyderabad, Telangana

SLA Breach Alerting System

- Reduced SLA breach-driven case escalations by 70-80% by architecting and deploying an enterprise-scale Agentic AI platform on Google ADK, Vertex AI Agent Engine, proactively flagging at-risk Google Cloud support cases before breach.
- Reduced average issue-resolution time by implementing a knowledge-grounded RAG pipeline with Vertex AI Vector Search, retrieval evaluation, and generated AI-driven actionable hard- and soft-risk recommendations for case owners.
- Handled zero PII-exposure incidents and blocked prompt-injection/jailbreak attempts across the agent pipeline by implementing Cloud DLP, encryption/decryption tool callbacks, and Responsible AI guardrails in production.
- Cut manual case-monitoring effort and response turnaround time by automating real-time Gmail and Google Chat notification workflows that surfaced critical risks, escalation signals, and next-best actions to case owners in near real time.
- Productionized the platform with CI/CD pipelines, fault-tolerant retry mechanisms, and cloud-native deployment practices, delivering a secure, scalable, and highly available AI solution for enterprise production environments.

Agentic AI Platform for Accounts Payable Team

- Streamlined complex invoice lifecycles, cutting manual processing time by roughly 40%, by designing stateful multi-agent workflows in LangGraph with dynamic routing, checkpointing, and conditional execution across document analysis, validation, risk, and payment-decision agents.
- Achieved ~95% structured-data extraction accuracy across diverse invoice formats by building a hybrid document-extraction pipeline (PyMuPDF, pdfplumber, LLM-based fallback parsing) paired with a vendor-history-driven risk-scoring model to flag fraud and anomalies before payment routing.
- Improved auditability and compliance on high-risk transactions by architecting configurable business-rule and Human-in-the-Loop (HITL) approval workflows, partnering with finance stakeholders to balance automation with manual review.

PROJECTS

BERT-Based Ticket Classification and AWS Deployment

NLP, Hugging Face, XGBoost, Docker, AWS SageMaker, FastAPI

- Improved ticket-classification accuracy to ~92% by building an end-to-end NLP system comparing TF-IDF + traditional ML baselines against a fine-tuned BERT model, using feature engineering and XGBoost optimization for model selection.
- Delivered a low-latency real-time prediction endpoint by containerizing the inference service with Docker and FastAPI, storing artifacts in Amazon S3, pushing images to Amazon ECR, and deploying via Amazon SageMaker.

SKILLS

Agentic AI Frameworks: LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, MCP, A2A Protocol

AI / ML & GenAI: Machine Learning, Deep Learning, NLP, Fine-Tuning, Gemini, GPT Models, RAG Pipelines, Vector Databases (Pinecone, FAISS, Vertex AI Vector Search), LLM Evaluation, Prompt Engineering, Hugging Face

MLOps & Observability: CI/CD, Model Monitoring, Tracing, Fault-Tolerant Deployment

Programming & Databases: Python, FastAPI, Flask, API/Tool Integration, React.js, SQL, BigQuery, Spanner DB

Cloud & Infrastructure: GCP Vertex AI, Gemini Enterprise, Guardrails, Cloud DLP, Cloud Run/Agent Engine, AWS Bedrock, SageMaker, S3, ECR

DevOps: Docker, CI/CD, Git

CERTIFICATIONS

Google Cloud Certified Professional Machine Learning Engineer • Google Cloud Certified Associate Cloud Engineer • Google Generative AI Leader Certification • Microsoft Certified Azure AI Fundamentals (AI-900) • AWS Certified AI Practitioner

ACHIEVEMENTS

- First Runner-Up, Virtusa Agentic AI Hackathon — Designed and shipped a full-stack application AI application using FastAPI, React.js, integrating AI-powered document and voice intelligence through Vision API-based OCR and Speech-to-Text capabilities.

EDUCATION

B.Tech, Computer Science & Engineering

Bonam Venkata Chalamayya Engineering College, Odalarevu

July 2019 – May 2023